

HAT-P-36b

R-0859

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_240331 · created 2026-07-28T21:43:58+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460400.7148 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.045
Transit depth (Rp/Rs)^2	2.45 +/- 1.41 [%]
Orbital Inclination (inc)	84.69 +/- 2.83
Airmass coefficient 1 (a1)	1.05 +/- 0.59
Airmass coefficient 2 (a2)	-0.22 +/- 0.48
Scatter in the residuals of the lightcurve fit is	52.83 %
Best Comparison Star	#1 - [44, 289]
Optimal Aperture	2.65
Optimal Annulus	7.0
Transit Duration (day)	0.0924 +/- 0.0091

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.045 vs 0.126 (deviation 0.67σ)
Duration fit vs published	0.0924 d vs 0.0925 d (deviation 0.01σ)
Mid-time O–C	12.1 min
Depth SNR	3.5
Residual scatter	52.83 %

Light curve

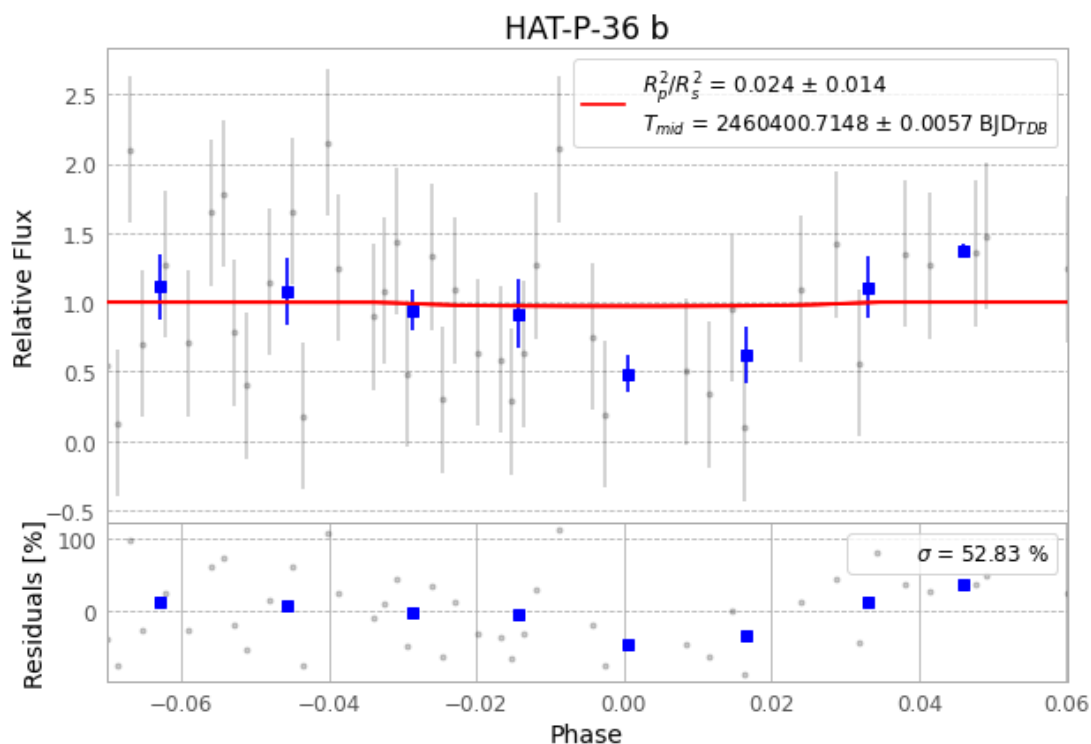


Figure 1 — FinalLightCurve_HAT-P-36 b_2024-03-30.png (SHA-256 57c5b0a843c1fead...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [222, 142], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b67db602858db381...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

85 FITS frames (56 MB), HATP-36240331024515.FITS → HATP-36240331065715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-240331.log (02d69f2d0866...), FinalLightCurve_HAT-P-36 b_2024-03-30.png (57c5b0a843c1...), FinalLightCurve_HAT-P-36 b_2024-03-30.csv (52799caf7979...)

Compute resources

Wall time 165.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 21:43Z.